

Affordable 3D printer for fun

da Vinci Mini is a compact and productive 3D printer that anybody can use. Weighing around 10kg, it measures just 39 x 33.5 x 36cm³ with a 15 x 15 x 15cm³ print bed. Its price, less than \$300, makes it even more attractive for designers, small-business product development teams, hobbyists, learners, casual home users and, even kids.

With a single-button print capability, da Vinci Mini is as easy to use as any normal desktop 2D printer. LEDs of different colours indicate the printer status, such as standby, printing and errors. The functionality is controlled through a PC, to which the printer is connected through USB or Wi-Fi. The printer has



da Vinci Mini is an affordable, compact and easy-to-use 3D printer (Image courtesy: XYZprinting)

on-board memory. So once you have uploaded the print designs, you can unplug the computer or move it out of the wireless home network without the printing getting interrupted. Wi-Fi connectivity allows sharing of the printer by multiple users.

The software's slicing precision and computing efficiency shorten the preparation time required before printing. Although the printing speed is slower than costlier 3D printers, it is pretty fast for a sub-\$300 model. The printer is eco-friendly and safe for use by the entire family as the filaments are made

using polylactic acid (PLA) plastics derived from corn starch, and are guaranteed to be free of heavy metals and di-(2-ethylhexyl)phthalate or DEHP. However, some users complain that da Vinci Mini works only with XYZ's filaments and no other generic brands.

Company: XYZprinting; **Country:** United States of America; **Website:** http://us.xyzprinting.com/us_en/Product/da-Vinci-Mini

Modular, customisable drone

The modular design trend has invaded the world of drones too. Makeblock has created a drone that can be assembled and programmed in minutes. Winner of Red Dot Award 2017, Airblock comprises magnetic parts that can be fitted together instantly to create various forms like hexacopter drone, triangle hovercraft, spider and spinner. You can even combine the blocks of this drone with other toys or blocks (like Lego) to build your own innovative automobiles and aircrafts! Made of a Styrofoam material, the drone does not damage walls or objects it dashes on. The blades are concealed within the blocks, so it is safe to fly even for kids.

If the assembly is as easy as saying snap, the drag-and-drop programming is even easier. Even those with no experience in programming can make their drone do amazing stunts! You can control Airblock using a smart device, customise its features including speed and colour of the LED, watch its movements, and get it racing with



Airblock is a set of modular blocks that let you build and program your own drone (Image courtesy: Makeblock)

another Airblock in 'versus mode.'

Airblock owes its effectiveness to the smart use of aerodynamic principles. According to the team, "The lift from the four propellers allows Airblock to do a number of stunts, and by selecting two propellers from either side, it can generate specific movements."

All that said, what people seem to love most about Airblock is that, for once, it is fun to watch your drone crash and break apart—because you can rebuild it and watch it fly again!

Company: Makeblock; **Country:** United States of America; **Website:** www.makeblock.com



Scan and edit 3D objects in real time using this advanced PC (Image courtesy: HP)

Computer with built-in 3D scanner

The Sprout Pro G2 computer by HP is a dream come true for designers as it can capture and render 3D objects in a jiffy. No clutter, no fuss, but the job gets done. You can scan a 3D object by just rotating it in front of the screen, edit the content in real time using a special pressure-sensitive active pen, and also create visual, hands-on experiences for collaboration, shopping, learning, etc, using the computer's dual multi-touch displays, unique 20-point Touch Mat and high-resolution cameras.

The high-resolution, full high definition (HD) projector projects on the 2.2mm thick horizontal Touch Mat. The mat's 20-point capacitive surface lets up to four hands work at the same time. It can work wonders together with the active pen, which has 2048 levels of pressure sensitivity to enable intuitive, precision digital inking. When you write on the Touch Mat with this pen, it feels as if you are writing on a paper with a pen.

Software tools like the WorkTools suite and HP 3D Scan Software Pro v5 help you harness the capabilities of the innovative hardware. You can capture objects in real time, create precise 360° 3D models using enhanced structured-light scanning technology, manipulate content in 2D and 3D through an interactive clipboard, manage how the displays appear on a third monitor, and so on. The Professional Software Development Kit (SDK) also gives you advanced hardware access to create apps for this specialised PC. For those tasked with 3D object handling, Sprout Pro is a product worth considering.

Company: HP; **Country:** United States of America; **Website:** <http://www8.hp.com/in/en/products/workstations/product-detail.html?oid=10912419>